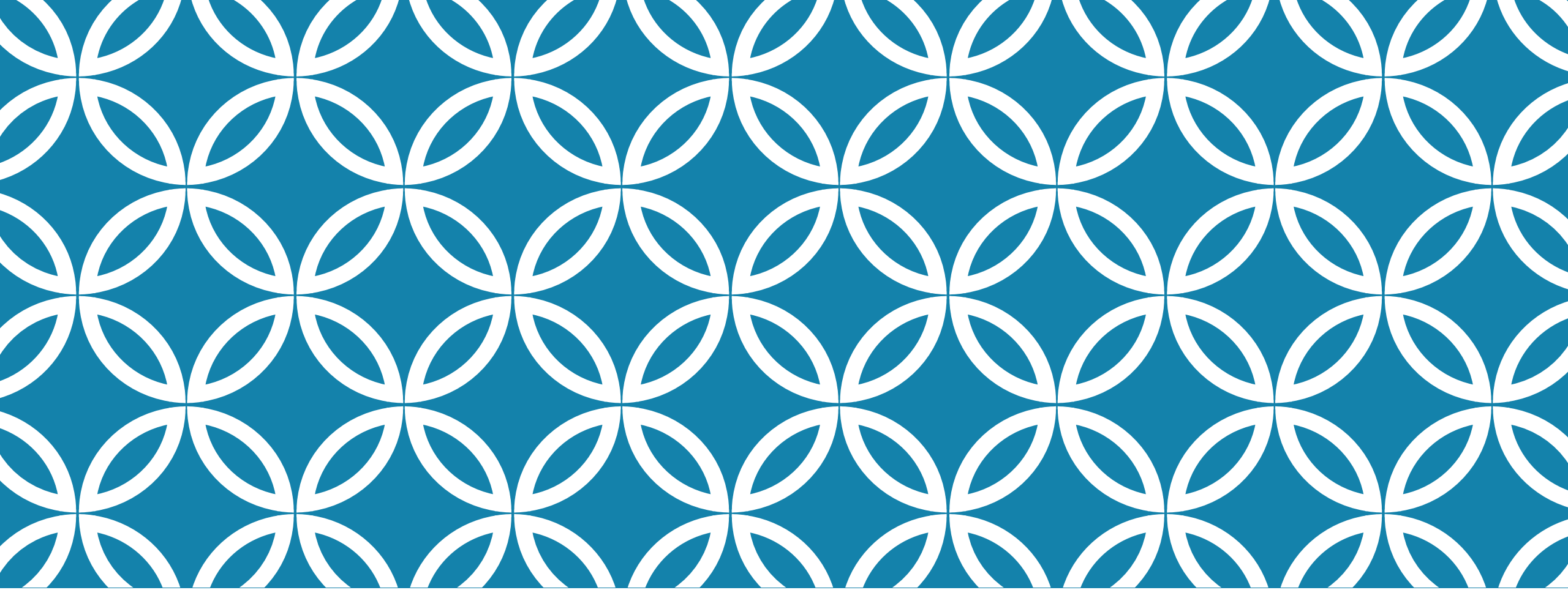


IMAGE FORMATION

FUNDAMENTALS OF IMAGE FORMATION

Image formation is an **analog to digital conversion** of an image with the help of 2D Sampling and Quantization techniques that is done by the capturing devices like cameras. In general, we see a 2D view of the 3D world.

In the same way, the formation of the analog image took place. It is basically a conversion of the 3D world that is our analog image to a 2D world that is our Digital image.

Generally, a frame grabber or a digitizer is used for sampling and quantizing the analog signals.

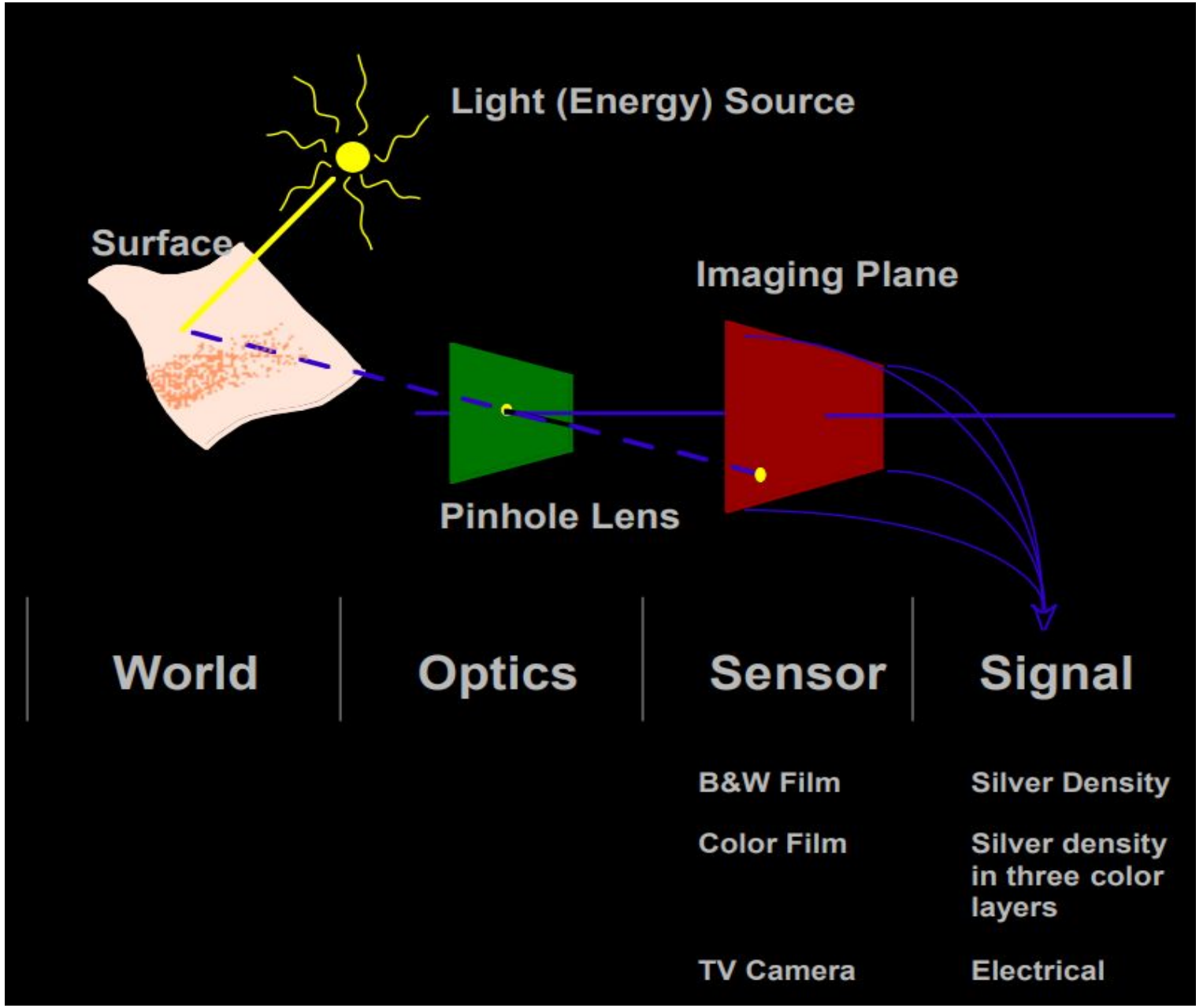
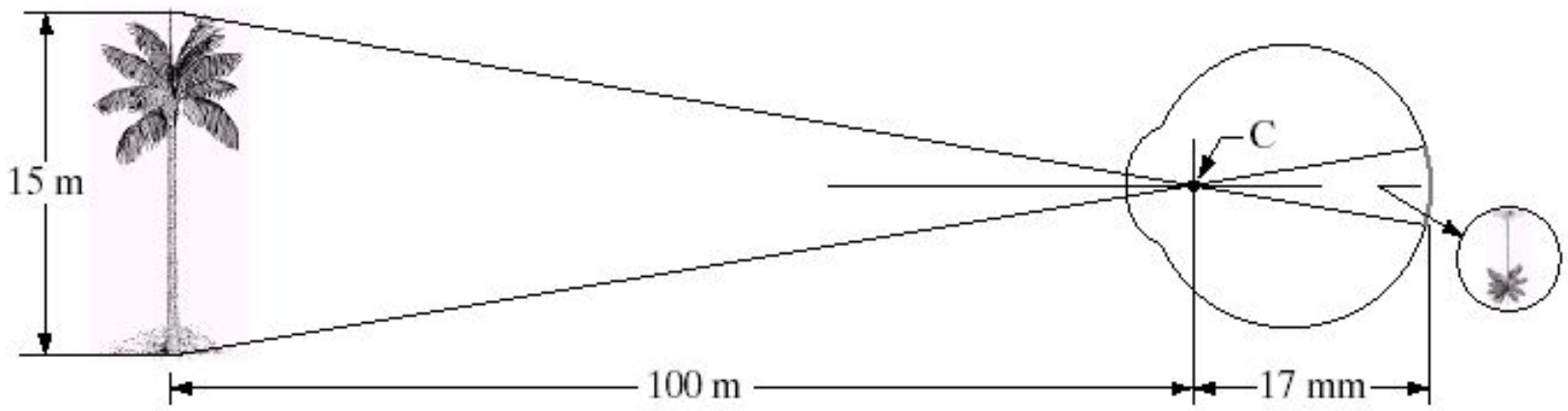
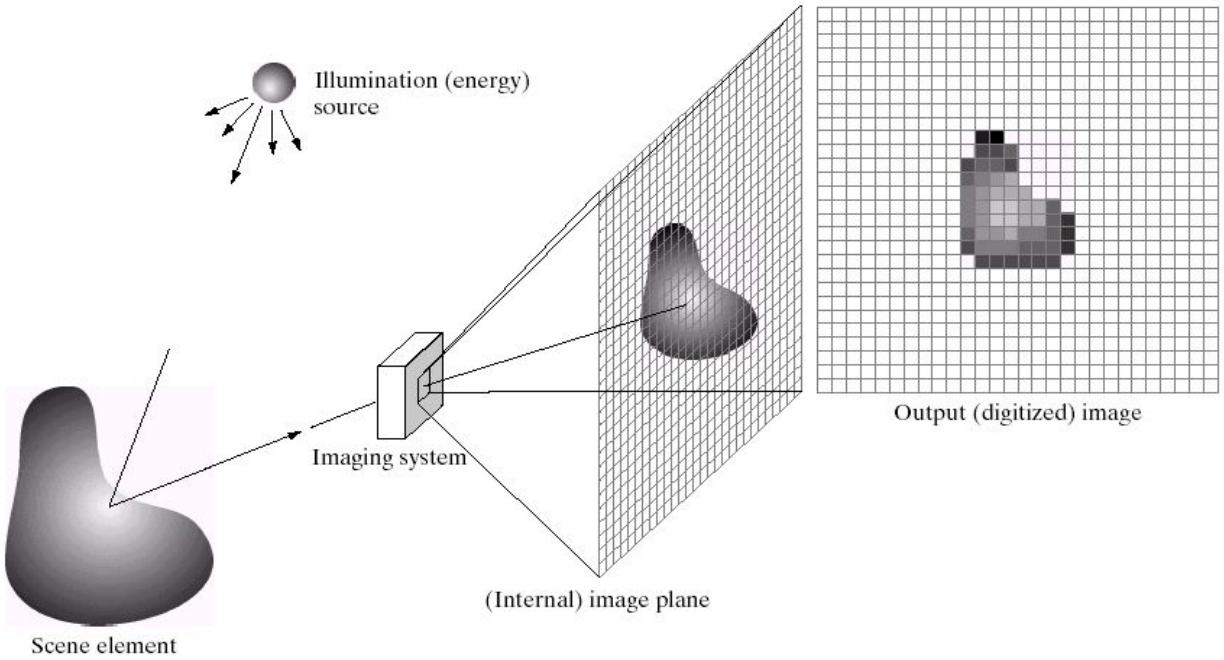

Light (Energy) Source
Surface
Imaging Plane
Pinhole Lens
World
Optics
Sensor
Signal
B&W Film
Silver Density
Color Film
Silver density in three color layers
TV Camera
Electrical

IMAGE FORMATION IN THE EYE

FIGURE 2.3
Graphical representation of the eye looking at a palm tree. Point *C* is the optical center of the lens.



SIMPLE IMAGE FORMATION MODEL



a
b c d e

FIGURE 2.15 An example of the digital image acquisition process. (a) Energy ("illumination") source. (b) An element of a scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

IMAGING

The mapping of a **3D world object into a 2D digital image plane is called *imaging***. In order to do so, each point on the 3D object must correspond to the image plane. We all know that light reflects from every object that we see thus enabling us to capture all those light-reflecting points in our image plane.

Various factors determine the quality of the image like spatial factors or the lens of the capturing device.

CONTINUATION

Optical Systems

The lenses and mirrors are crucial in focusing the light coming from the 3D scene to produce the image on the image plane. These systems define ***how light is collected and where it is directed and consequently affects the sharpness and quality of the image produced.***

Image Sensors

The goals of image sensors like the ***CCD or the CMOS sensors*** are to simply transform the optical image into an electronic signal. These sensors differ by sensitivity, the resolution that they deliver affecting the image as a whole.

CONTINUATION

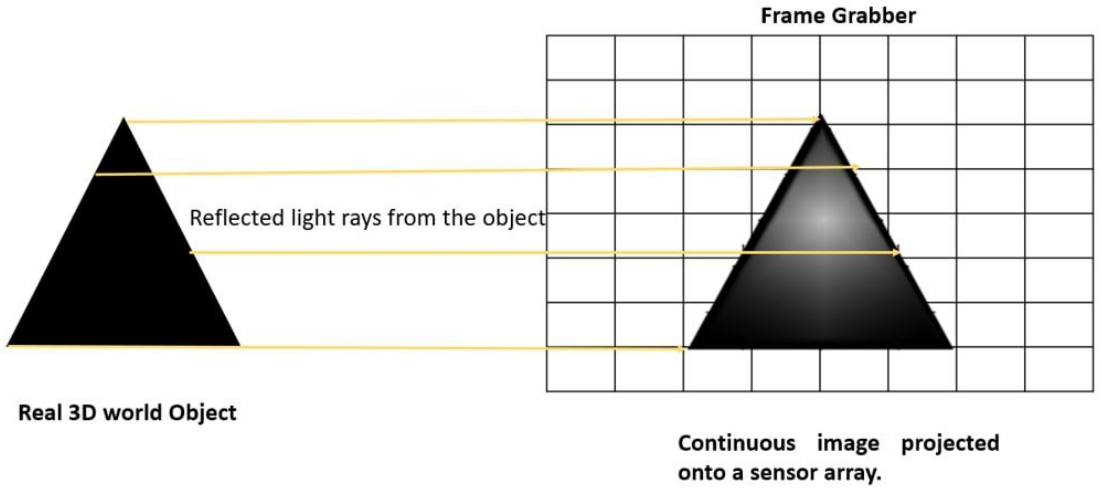
Resolution and Sampling

Resolution is defined as the ***sharpness of an image and it occurs technically as the number of pixels an image can hold.***

Sampling is the act of ***taking samples or discretizing a digital signal and representing a continuous analog signal*** as a grouping of discrete values. It can be seen that higher resolution and appropriative sampling rates are required in order to provide detailed and accurate images.

IMAGE PROCESSING

Image processing can be described as act of modifying and enhancing digital images by using algorithms. Pre-processing includes activities like ***filtering, noise reduction and color correction*** that enhance image quality and information extraction



COLOR AND PIXELATION

In digital Imaging, a frame grabber is placed at the image plane which is like a sensor. It aims to focus the light on it and the continuous image is pixelated via the reflected light by the 3D object. The light that is focused on the sensor generates an electronic signal.

Each pixel that is formed may be colored or grey depending on the intensity of the sampling and quantization of the light that is reflected and the electronic signal that is generated via them.

All these pixels form a digital image. The density of these pixels determines the image quality. The more the density the more the clear and high-resolution image we will get.

FORMING DIGITAL IMAGE

In order to form or create an image that is digital in nature, we need to have a ***continuous conversion of data into a digital form***. Thus, we require two main steps to do so:

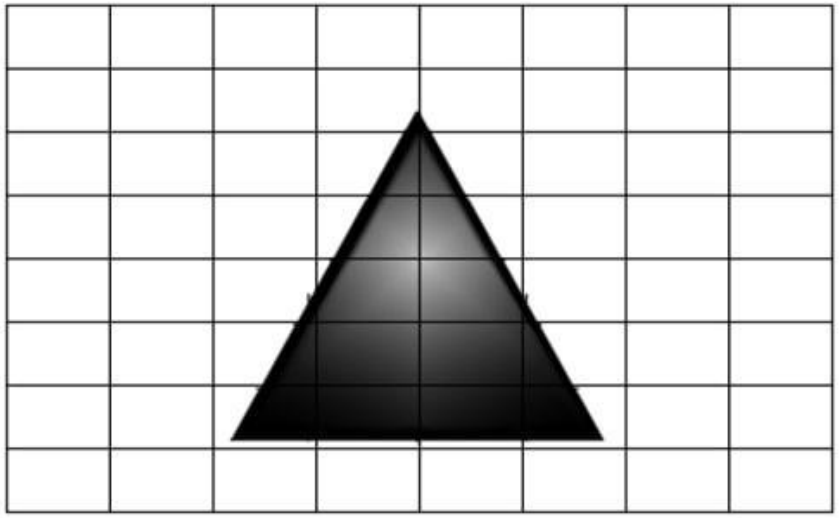
Sampling (2D): ***Sampling is a spatial resolution of the digital image***. And the rate of sampling determines the quality of the digitized image. The magnitude of the sampled image is determined as a value in image processing. It is related to the coordinates values of the image.

CONTINUATION

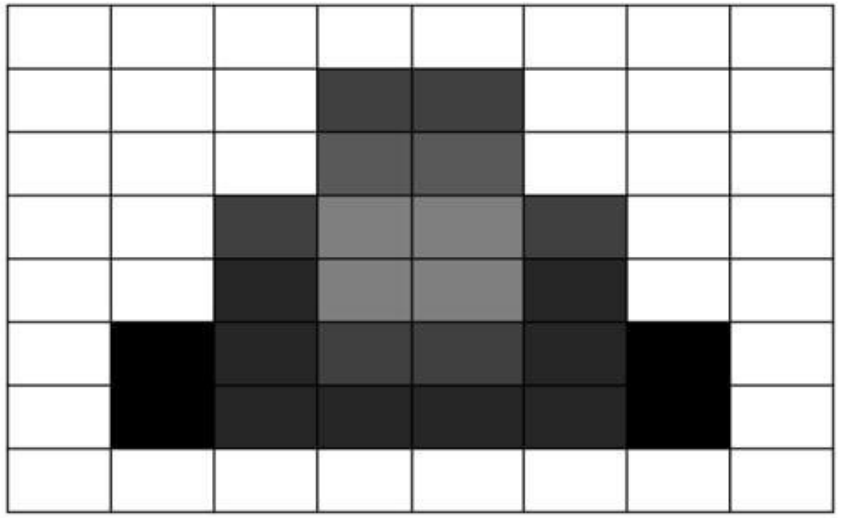
Quantization: *Quantization is the number of grey levels in the digital image*. The transition of the continuous values from the image function to its digital equivalent is called quantization. It is related to the intensity values of the image.

The normal human being acquires a high level of quantization levels to get the fine shading details of the image. ***The more quantization levels will result in the more clear image.***

CONTINUATION

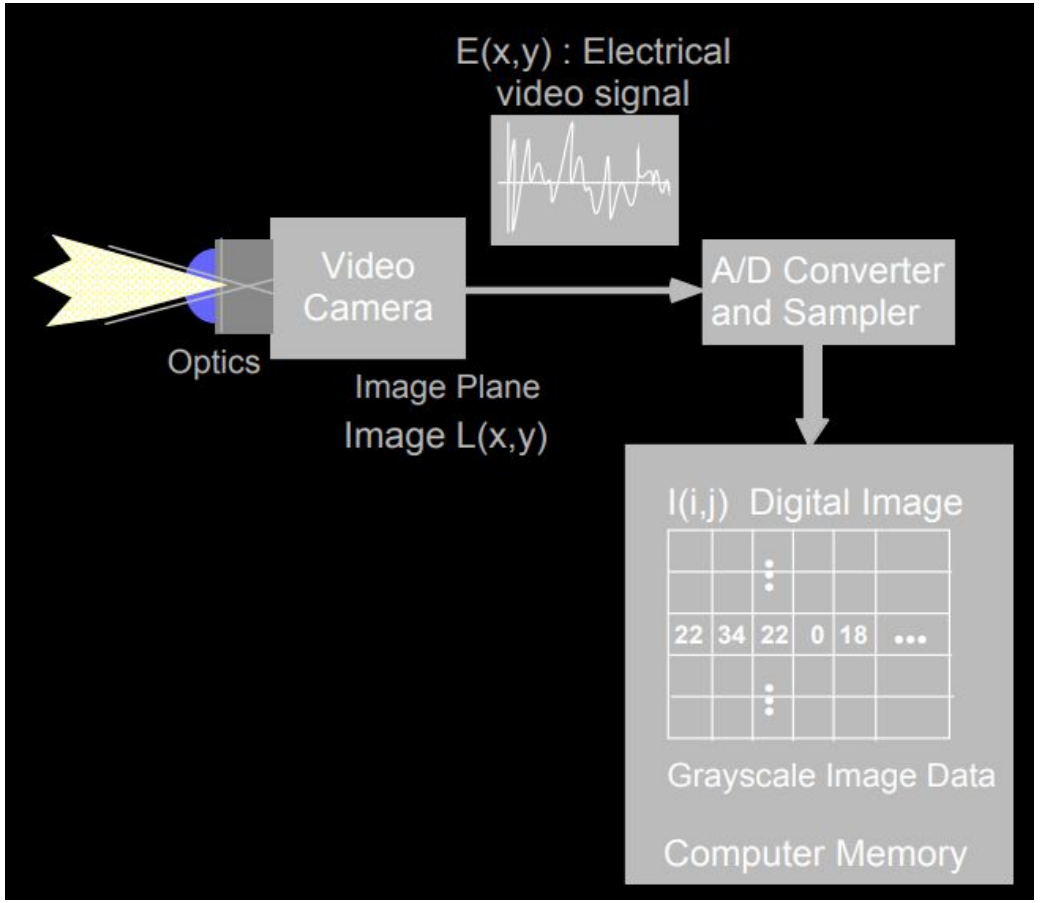


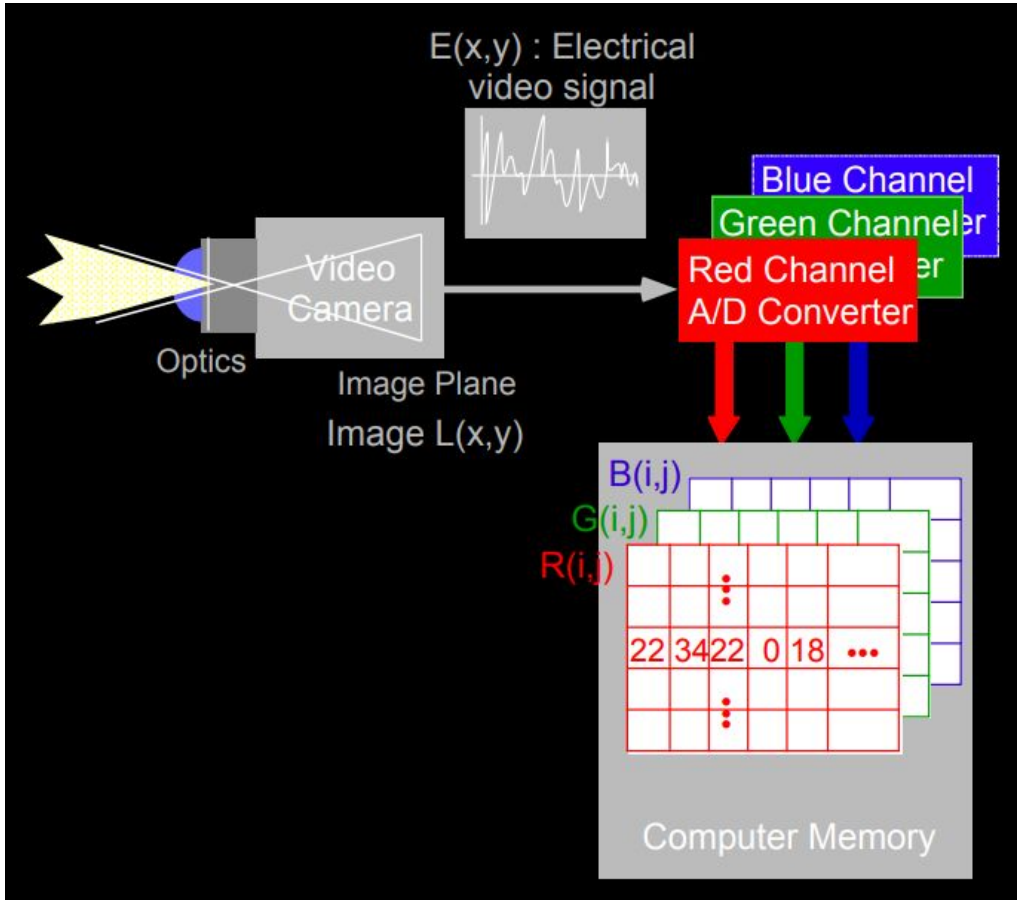
Continuous image projected onto a sensor array.



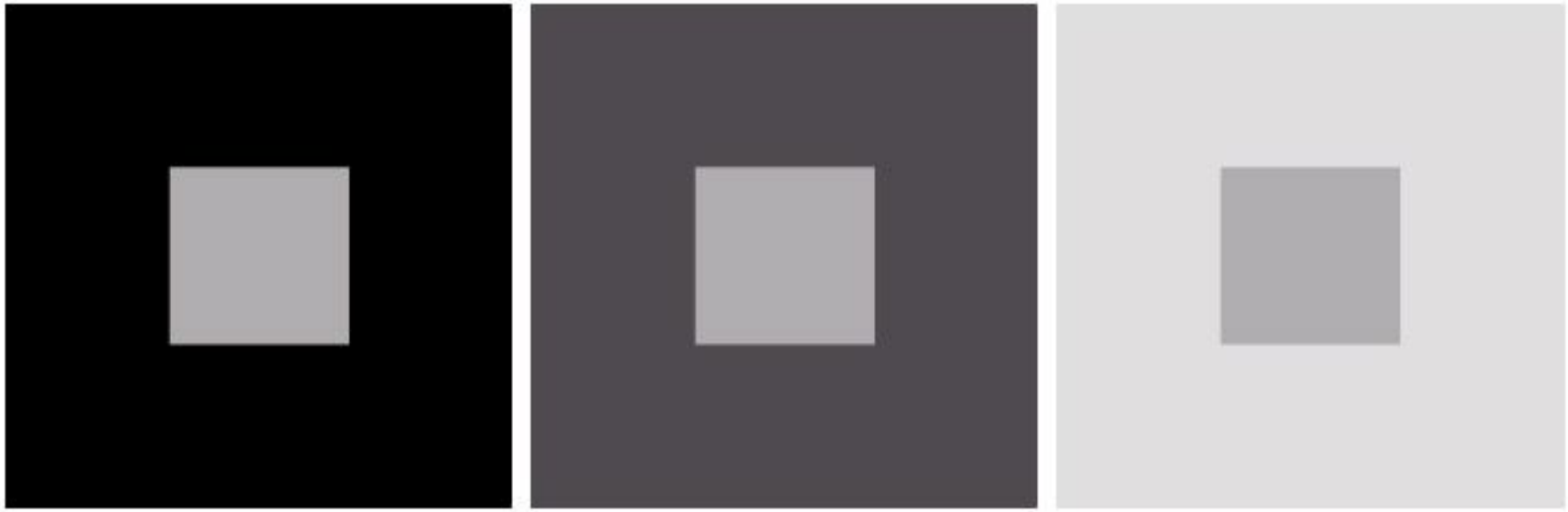
Result of image sampling and quantization.

CONTINUATION





SIMULTANEOUS CONTRAST



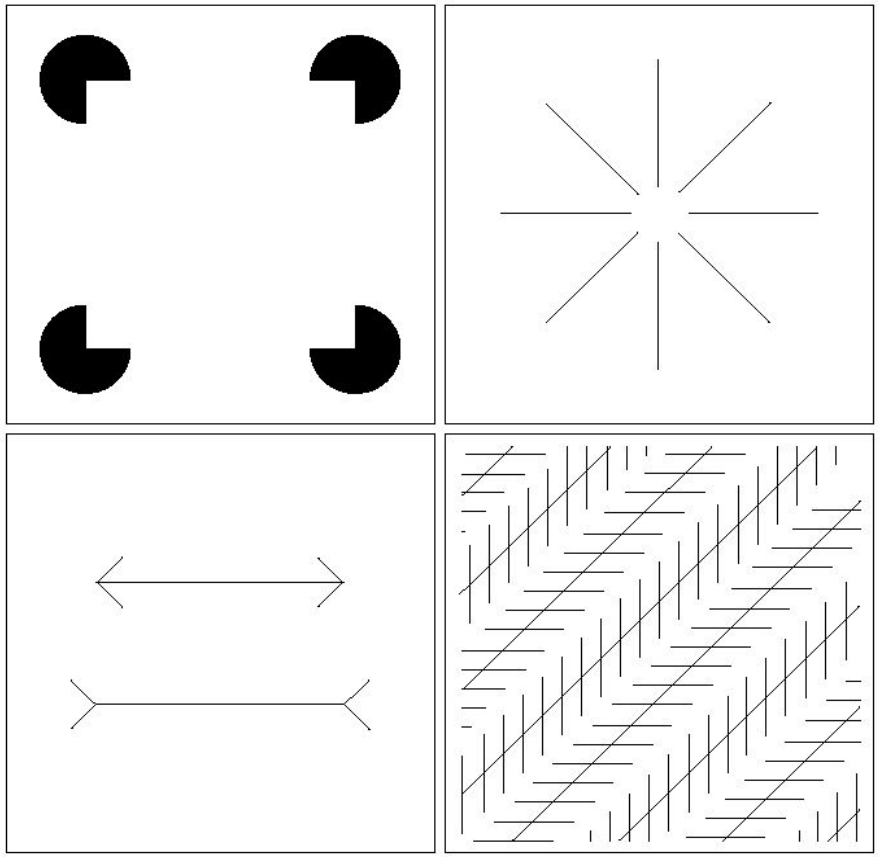
a b c

FIGURE 2.8 Examples of simultaneous contrast. All the inner squares have the same intensity, but they appear progressively darker as the background becomes lighter.

OPTICAL ILLUSION

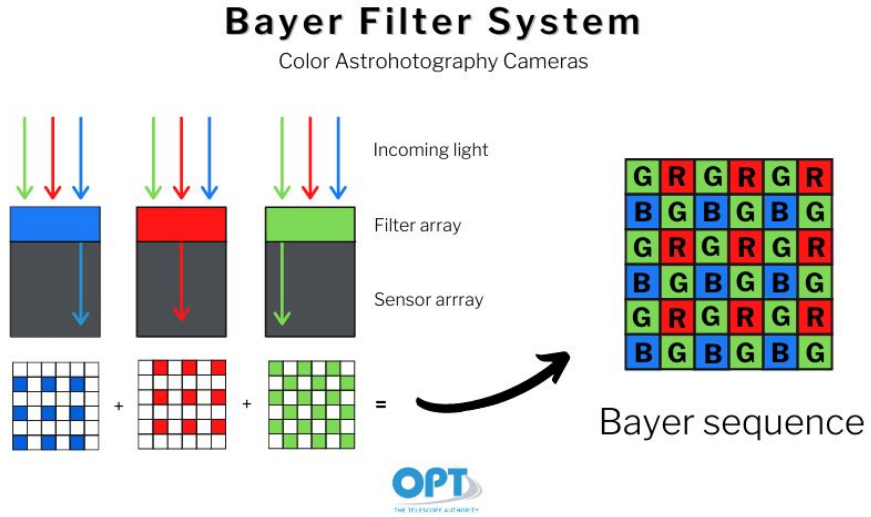
a b
c d

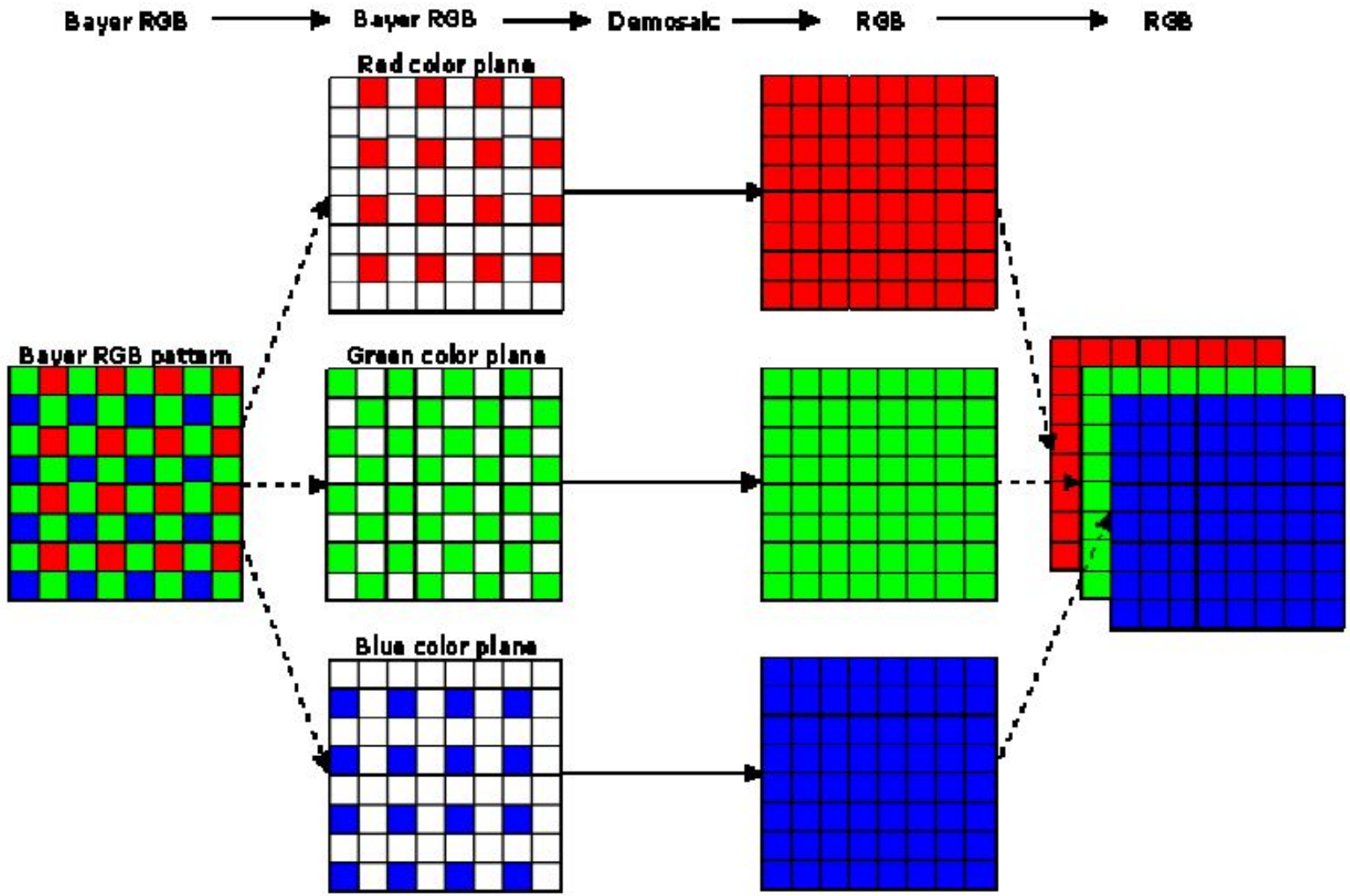
FIGURE 2.9 Some well-known optical illusions.



BAYER FILTER

A Bayer filter is a microscopic color filter array (CFA) placed over a digital camera's image sensor to capture color. Invented by Bryce Bayer at Kodak, it arranges filters in a repeating 2 × 2 grid with one red, one blue, and two green filters. [1, 2, 3]




Bayer RGB
Bayer RGB
Demosaic
RGB
RGB
Red color plane
Bayer RGB pattern
Green color plane
Blue color plane

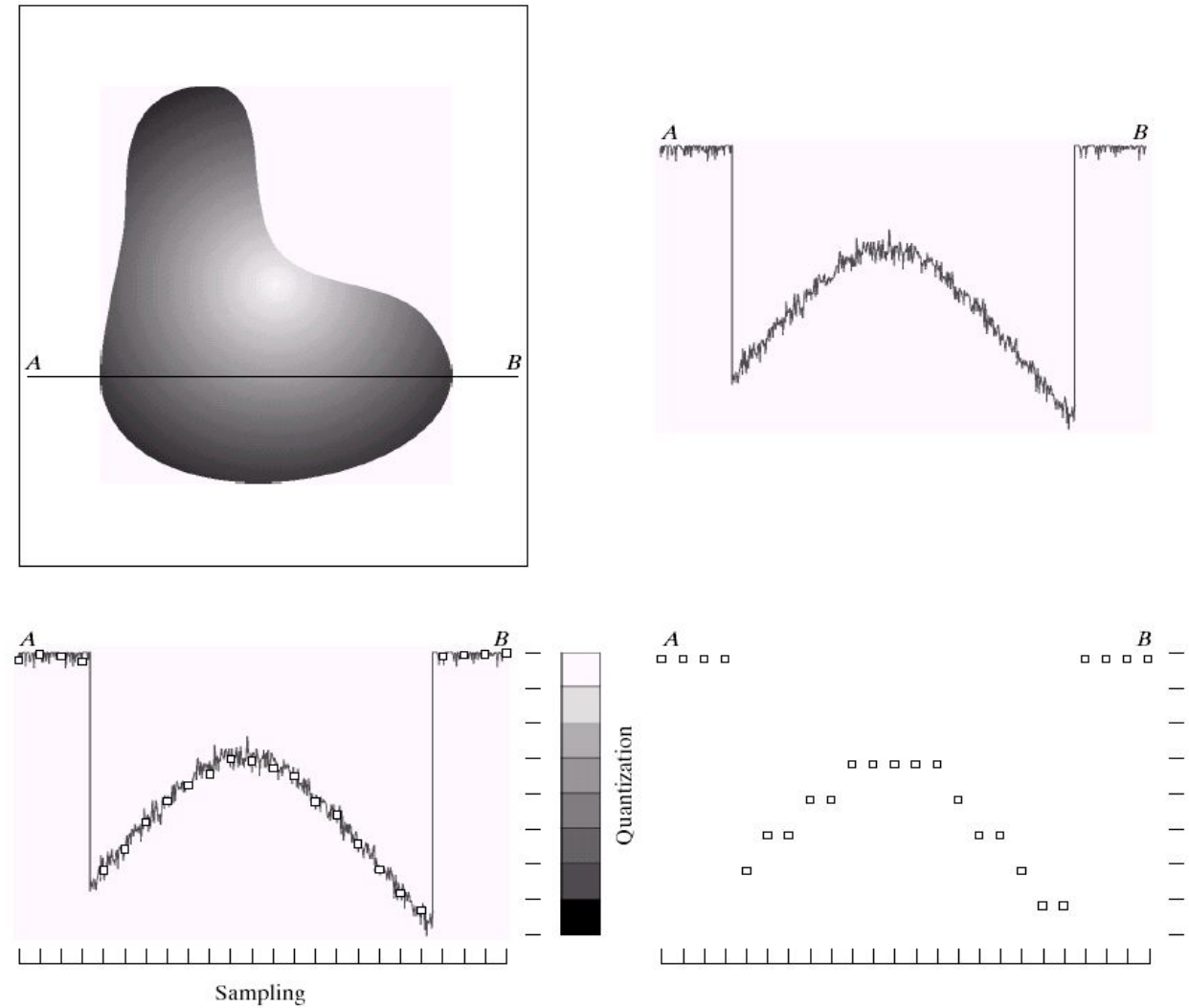
SAMPLING STRATEGIES:

- Spatial sampling
  - How many pixels?
  - What arrangement of pixels?
- Brightness sampling
  - How many brightness values?
  - Spacing of brightness values?
- For video, also the question of time sampling.

QUANTIZATION

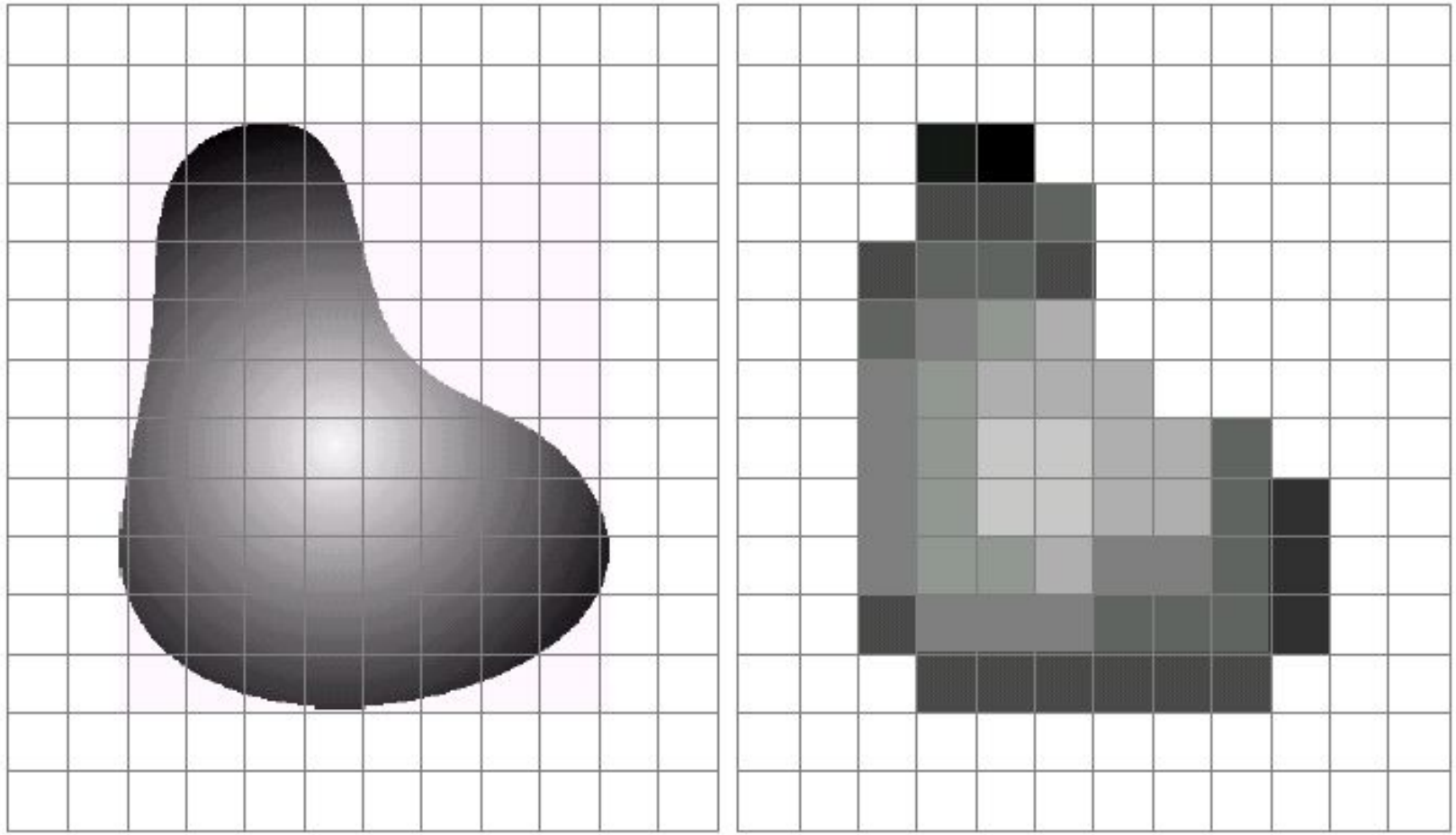
I(x,y) = continuous signal: $0 \leq I \leq M$

- Want to quantize to K values 0,1,....K-1
- K usually chosen to be a power of 2:
- Mapping from input signal to output signal is to be determined.
- Several types of mappings: uniform, logarithmic, etc



a b
c d

FIGURE 2.16 Generating a digital image. (a) Continuous image. (b) A scan line from *A* to *B* in the continuous image, used to illustrate the concepts of sampling and quantization. (c) Sampling and quantization. (d) Digital scan line.

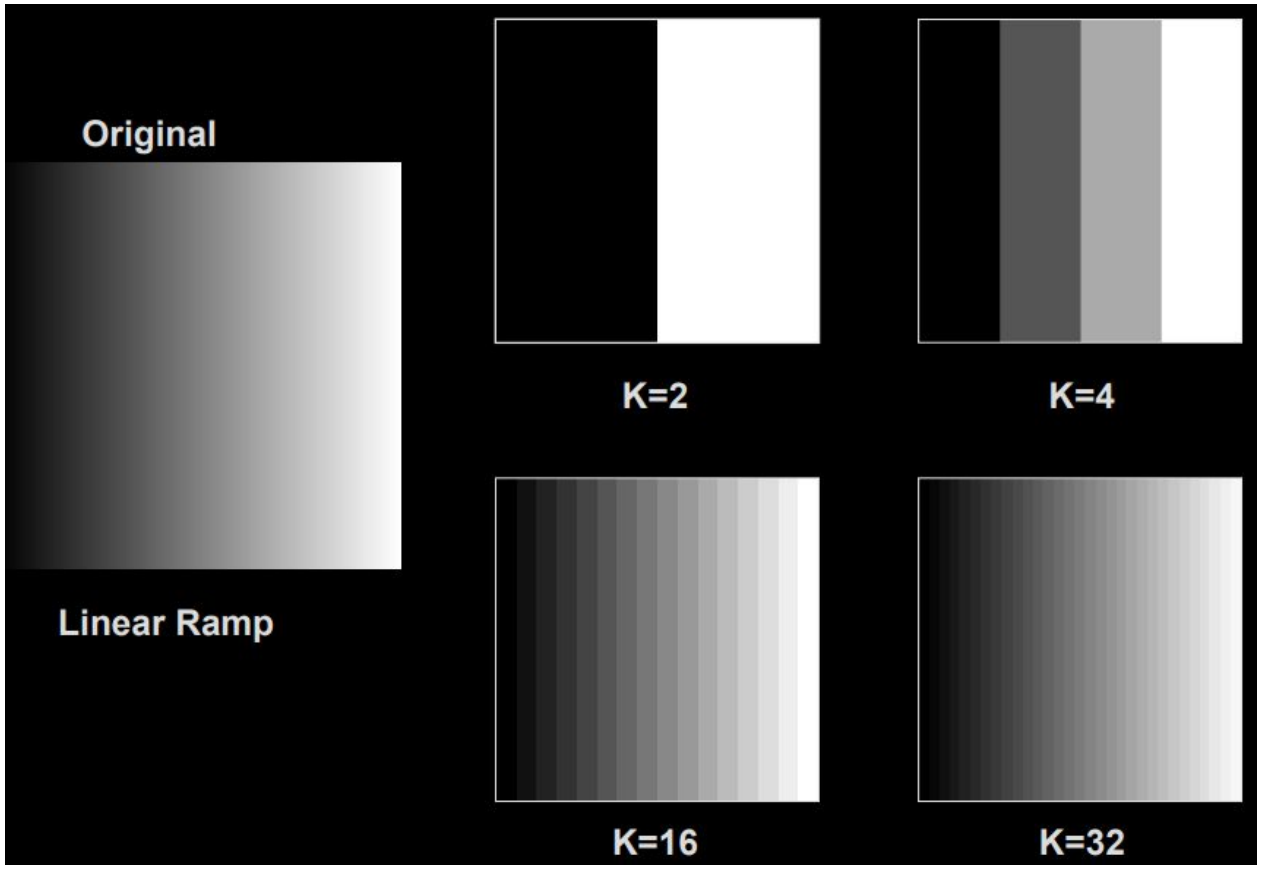


a b

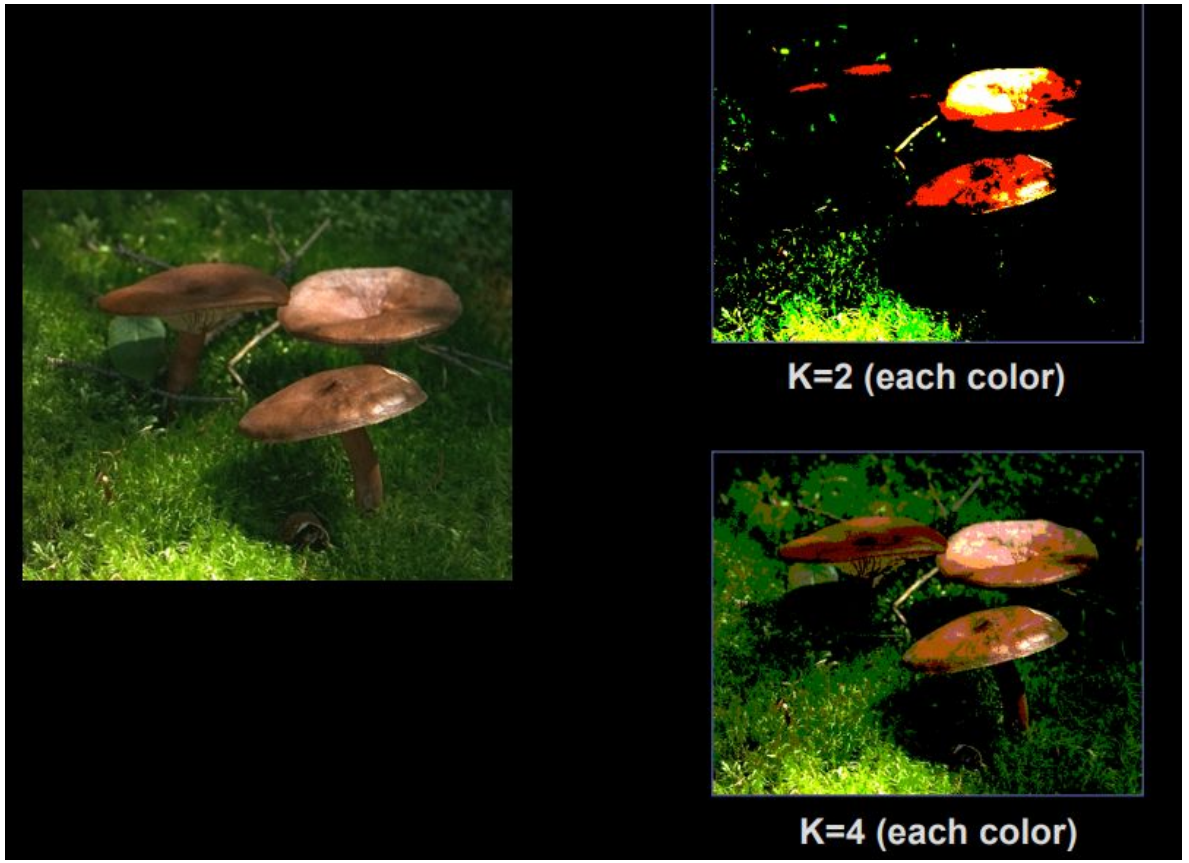
FIGURE 2.17 (a) Continuos image projected onto a sensor array. (b) Result of image sampling and quantization.

CONTINUATION

K: #Levels	#Bits
2	1
4	2
8	3
16	4
32	5
64	6
128	7
256	8

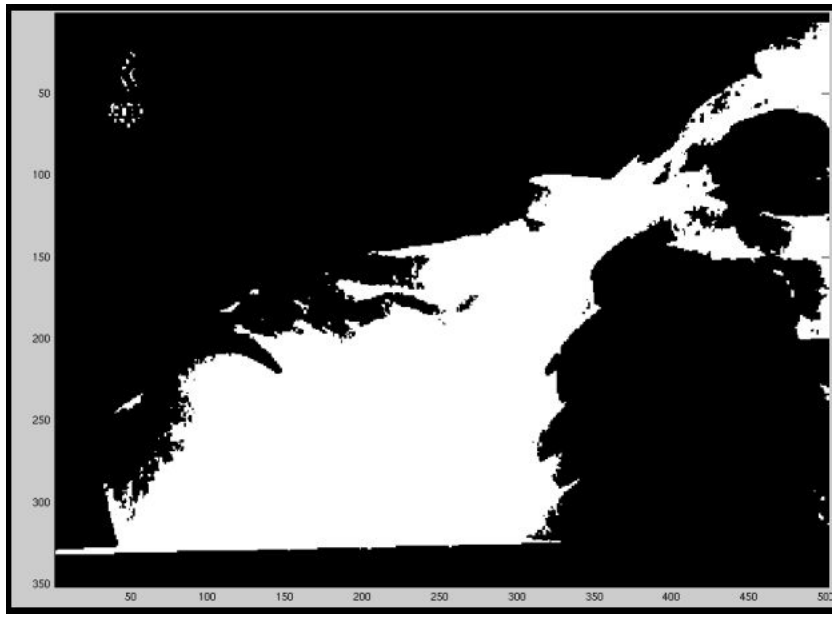


CONTINUATION

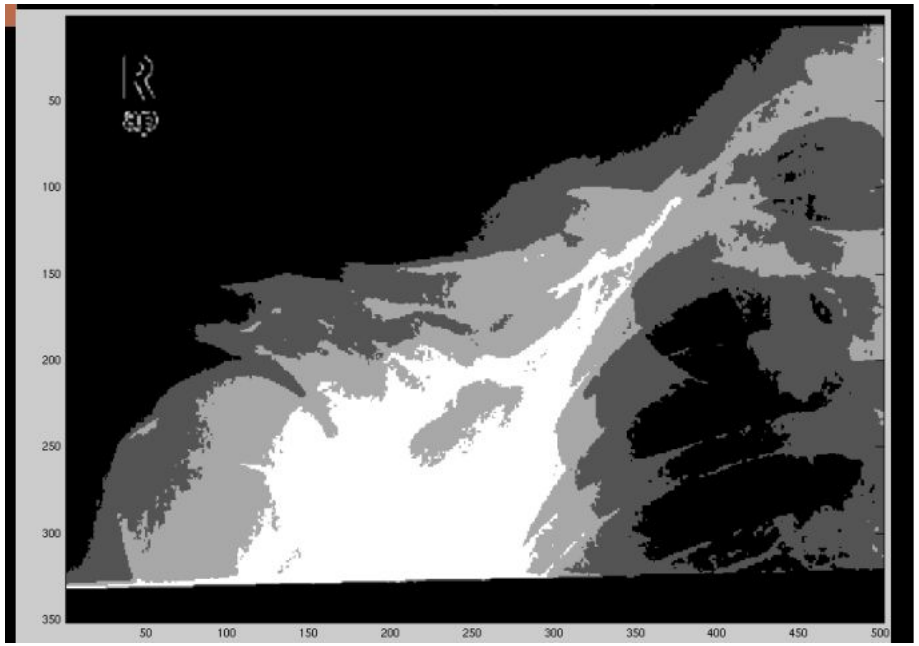


CONTINUATION

Digital X-Rays – 1 Bit

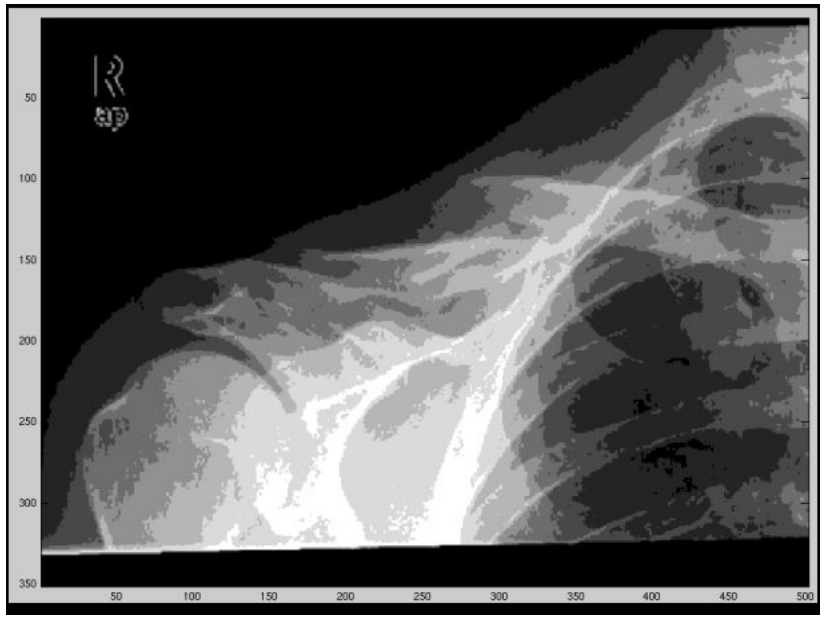


Digital X-Rays – 2 Bits

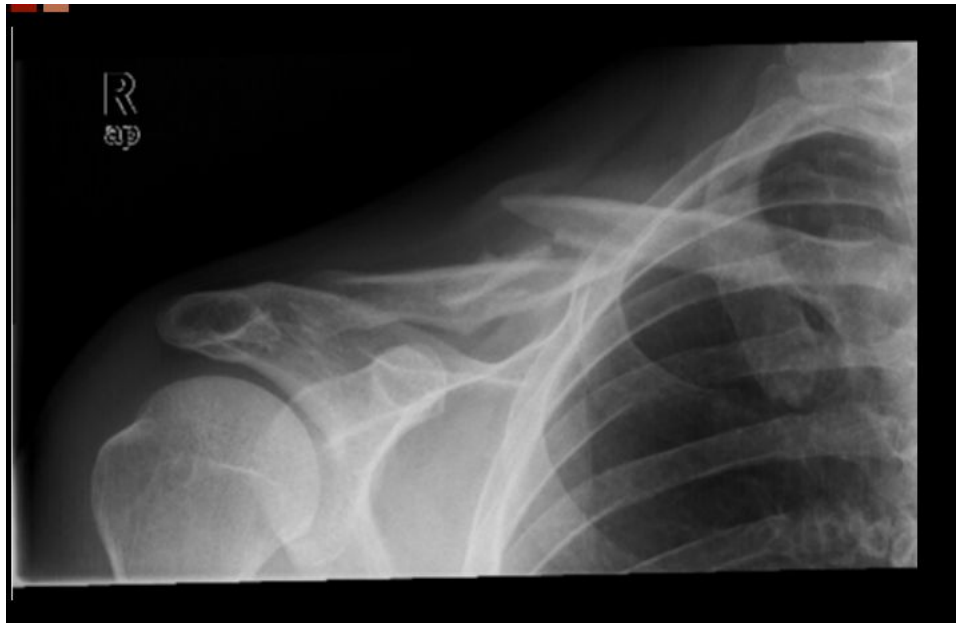


CONTINUATION

Digital X-Rays – 3 Bits



Digital X-Rays – 8 Bits



REPRESENTATION OF DIGITAL IMAGES

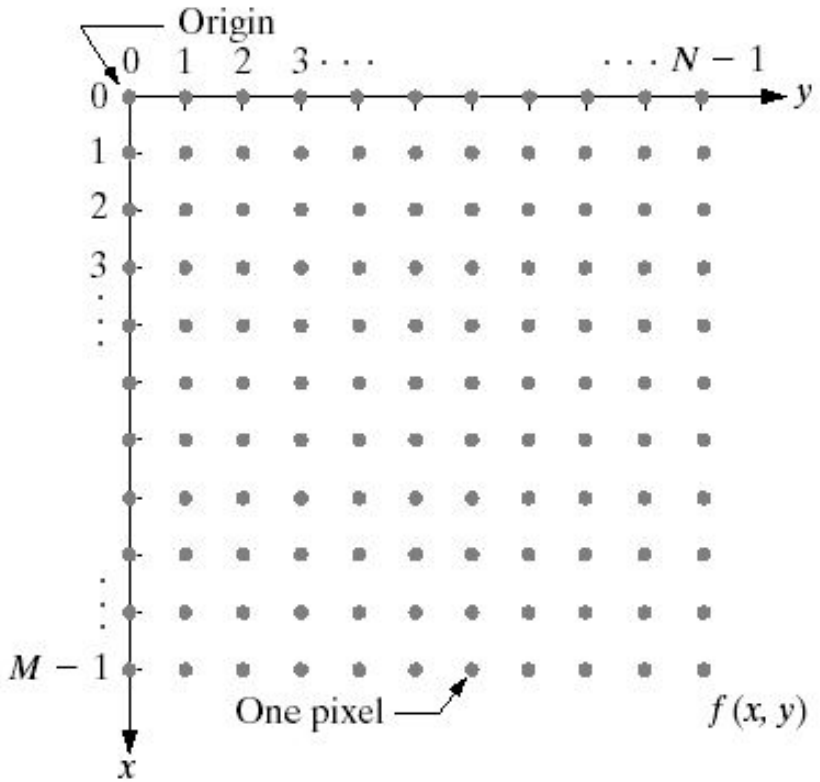


FIGURE 2.18
Coordinate convention used in this book to represent digital images.

NO. OF STORAGE BITS

TABLE 2.1
Number of storage bits for various values of N and k.

N/k	1 ($L = 2$)	2 ($L = 4$)	3 ($L = 8$)	4 ($L = 16$)	5 ($L = 32$)	6 ($L = 64$)	7 ($L = 128$)	8 ($L = 256$)
32	1,024	2,048	3,072	4,096	5,120	6,144	7,168	8,192
64	4,096	8,192	12,288	16,384	20,480	24,576	28,672	32,768
128	16,384	32,768	49,152	65,536	81,920	98,304	114,688	131,072
256	65,536	131,072	196,608	262,144	327,680	393,216	458,752	524,288
512	262,144	524,288	786,432	1,048,576	1,310,720	1,572,864	1,835,008	2,097,152
1024	1,048,576	2,097,152	3,145,728	4,194,304	5,242,880	6,291,456	7,340,032	8,388,608
2048	4,194,304	8,388,608	12,582,912	16,777,216	20,971,520	25,165,824	29,369,128	33,554,432
4096	16,777,216	33,554,432	50,331,648	67,108,864	83,886,080	100,663,296	117,440,512	134,217,728
8192	67,108,864	134,217,728	201,326,592	268,435,456	335,544,320	402,653,184	469,762,048	536,870,912

SPATIAL GRAY-LEVEL RESOLUTION



FIGURE 2.19 A 1024 × 1024, 8-bit image subsampled down to size 32 × 32 pixels. The number of allowable gray levels was kept at 256.

BASIC RELATIONSHIP BETWEEN PIXELS

Neighbors of a pixel

$N_4(p)$: 4-neighbors of p

$(x+1, y)$ $(x-1, y)$ $(x, y+1)$ $(x, y-1)$

$N_D(p)$: 4-diagnol neighbors of p

$(x+1, y+1)$ $(x+1, y-1)$ $(x-1, y-1)$ $(x-1, y+1)$

$N_8(p)$: 8-neighbors of p

$N_4(p)$ $N_D(p)$

CONTINUATION

Adjacency

V : The set of gray-level values used to define adjacency

4-adjacency: Two pixels p and q with values from V are

4-adjacency if q is in the set $N_4(p)$

8-adjacency: Two pixels p and q with values from V are

8-adjacency if q is in the set $N_8(p)$

IMAGE TRANSFORMATION

The fundamentals of image formation involve projecting a 3D scene onto a 2D image plane using optics (like pinhole models).

Image transformations are geometric operations applied to pixel coordinates to change the shape, position, or orientation of an image.

These geometric transformations map a point (x, y) to a new point (x', y') and are organized into a hierarchy based on the properties they preserve:

ORTHOGONAL

Definition: A transformation that preserves vector lengths and dot products.

Key Matrix Property: Matrix R is orthogonal if its transpose equals its inverse: $R^T R = I$.

Determinant Constraint: $\det(R) = \pm 1$. A value of +1 yields standard rotations; -1 results in reflections.

Homogeneous Structure:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & 0 \\ r_{21} & r_{22} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Invariants: Preserves distances, angles, area, and structural hand orientation (if +1). Slideshare +3

EUCLIDIAN

Definition: Combines an orthogonal rotation matrix with a spatial shift vector.

Physical Intuition: Models rigid objects moving through space without being compressed, stretched, or warped.

Matrix Formulation:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & t_x \\ \sin\theta & \cos\theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Degrees of Freedom (DoF): **3 DoF** total (1 angle parameter θ, 2 movement offsets t_x, t_y).

Invariants: Strictly preserves total area, distances between points, and internal angles. Slideshare +4

SIMILARITY

Definition: An extension of rigid body movement that adds uniform isotropic scaling factor s.

Matrix Formulation:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s\cos\theta & -s\sin\theta & t_x \\ s\sin\theta & s\cos\theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Degrees of Freedom: **4 DoF** (1 scale factor s, 1 rotation angle θ, 2 translation offsets t_x, t_y).

Invariants: Preserves angles, parallel line relationships, and length ratios. Absolute area scales by s^2. SlideServe +1

AFFLINE

Definition: A linear transformation followed by a translation step.

New Behaviors: Non-uniform scaling (independent x and y adjustments) along with shearing transformations.

Matrix Formulation: Any inversion-safe 3 × 3 matrix where the bottom row is strictly restricted to $[0, 0, 1]$:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Degrees of Freedom: **6 DoF** (4 linear warping values, 2 translation shifts).

Invariants: Keeps straight lines straight and parallel lines parallel. Preserves parallel area ratios. Slideshare +4

PROJECTIVE

Definition: The most general planar mapping option under linear homogeneous frameworks.

Geometric Behavior: Models changes in viewpoint where the camera tilts relative to a scene, mapping squares into arbitrary quadrilaterals.

Matrix Formulation: The bottom row constraint is removed, opening all parameters to scaling fluctuations:

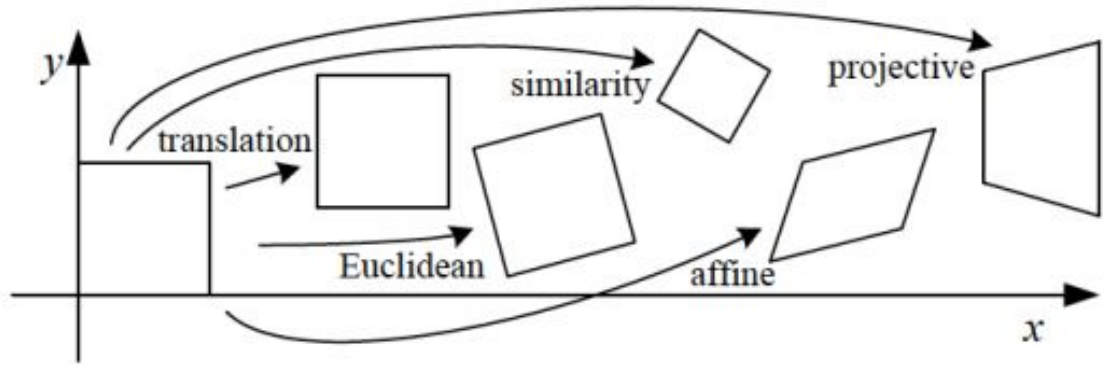
$$\begin{bmatrix} \tilde{x}' \\ \tilde{y}' \\ w' \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Degrees of Freedom: **8 DoF** (9 parameters total, defined up to an arbitrary global scale factor).

Invariants: Straight lines remain straight, but parallel lines converge toward horizons. Preserves the cross-ratio of points. Slideshare +4

Transformation Name	Matrix Constraints	Degrees of Freedom	Invariant Attributes
Orthogonal	$R^T R = I, \det(R) = \pm 1$, no translations	**1 to 3**	Distances, Angles, Area
Euclidean (Rigid)	Top-left is orthogonal, bottom row is $[0, 0, 1]$	**3** (1 rotation, 2 translations)	Distances, Angles, Area
Similarity	Top-left is scaled orthogonal, bottom row is $[0, 0, 1]$	**4** (1 scale, 1 rotation, 2 translations)	Angles, Length Ratios
Affine	No restrictions on top-left, bottom row is $[0, 0, 1]$	**6** (4 linear warp, 2 translations)	Parallelism, Parallel Area Ratios
Projective	Fully unconstrained 3 × 3 matrix	**8** (Normalized matrix scale)	Straight Lines, Cross-Ratios

Hierarchy of 2D Planar Transformations



Transformation	Matrix	# DoF	Preserves	Icon
translation	$\begin{bmatrix} \boldsymbol{I} \mid \boldsymbol{t} \end{bmatrix}_{2\times3}$	2	orientation	
rigid (Euclidean)	$\begin{bmatrix} \boldsymbol{R} \mid \boldsymbol{t} \end{bmatrix}_{2\times3}$	3	lengths	
similarity	$\begin{bmatrix} s\boldsymbol{R} \mid \boldsymbol{t} \end{bmatrix}_{2\times3}$	4	angles	
affine	$\begin{bmatrix} \boldsymbol{A} \end{bmatrix}_{2\times3}$	6	parallelism	
projective	$\begin{bmatrix} \tilde{\boldsymbol{H}} \end{bmatrix}_{3\times3}$	8	straight lines	

PROJECTIVE GEOMETRY

	Euclidean	similarity	affine	projective
Transformations				
rotation	X	X	X	X
translation	X	X	X	X
uniform scaling		X	X	X
nonuniform scaling			X	X
shear			X	X
perspective projection				X
composition of projections				X
Invariants				
length	X			
angle	X	X		
ratio of lengths	X	X		
parallelism	X	X	X	
incidence	X	X	X	X
cross ratio	X	X	X	X

PYTHON CODE

```python
import cv2
import numpy as np
import matplotlib.pyplot as plt

def create_synthetic_image():
    """Generates a simple 400x400 checkerboard image for visualization."""
    img = np.zeros((400, 400, 3), dtype=np.uint8) + 255
    # Draw a distinct dark square grid
    cv2.rectangle(img, (100, 100), (300, 300), (40, 40, 40), -1)
    cv2.rectangle(img, (130, 130), (270, 270), (200, 200, 200), -1)
    # Add an asymmetrical corner anchor to track orientation/reflection
    cv2.circle(img, (130, 130), 15, (0, 0, 255), -1)
    return img

# Load the source asset
img = create_synthetic_image()
rows, cols, ch = img.shape
```

CONTINUATION

```
# ========================================
# 1. ORTHOGONAL TRANSFORMATION (Pure Rotation)
# ========================================
# Scale factor must be exactly 1.0 to preserve distances (orthogonal property)
angle_rad = np.radians(30)
cos_a, sin_a = np.cos(angle_rad), np.sin(angle_rad)

# Matrix: [cos(theta), -sin(theta), 0], [sin(theta), cos(theta), 0]
M_orthogonal = np.array([
    [cos_a, -sin_a, 0],
    [sin_a,  cos_a, 0]
], dtype=np.float32)

img_orthogonal = cv2.warpAffine(img, M_orthogonal, (cols, rows), borderValue=(255, 255, 255))
```

CONTINUATION

```
# ========================================
# 2. EUCLIDEAN TRANSFORMATION (Rigid: Rot + Trans)
# ========================================
# Compute rotation matrix around center point, keeping scale = 1.0
center = (cols // 2, rows // 2)
M_rigid = cv2.getRotationMatrix2D(center, angle=45, scale=1.0)
# Append translation offsets directly into the third column
M_rigid[0, 2] += 50  # tx offset
M_rigid[1, 2] -= 30  # ty offset

img_euclidean = cv2.warpAffine(img, M_rigid, (cols, rows), borderValue=(255, 255, 255))
```

CONTINUATION

```
# ==========================================
# 3. AFFINE TRANSFORMATION (Parallelism Preserved)
# ==========================================
# Maps 3 source coordinates to 3 target coordinates to allow shear/scaling
pts_src_affine = np.float32([[50, 50], [200, 50], [50, 200]])
pts_dst_affine = np.float32([[70, 60], [220, 40], [40, 230]])

M_affine = cv2.getTransformationMatrix = cv2.getAffineTransform(pts_src_affine, pts_dst_affine)
img_affine = cv2.warpAffine(img, M_affine, (cols, rows), borderValue=(255, 255, 255))
```

CONTINUATION

```
# ========================================
# 4. PROJECTIVE TRANSFORMATION (Homography)
# ========================================
# Maps 4 source points to 4 target points (lines converge to vanishing points)
pts_src_proj = np.float32([[0, 0], [cols-1, 0], [0, rows-1], [cols-1, rows-1]])
pts_dst_proj = np.float32([[60, 80], [cols-80, 40], [20, rows-40], [cols-40, rows-80]])

M_projective = cv2.getPerspectiveTransform(pts_src_proj, pts_dst_proj)
img_projective = cv2.warpPerspective(img, M_projective, (cols, rows), borderValue=(255, 255, 255))
```

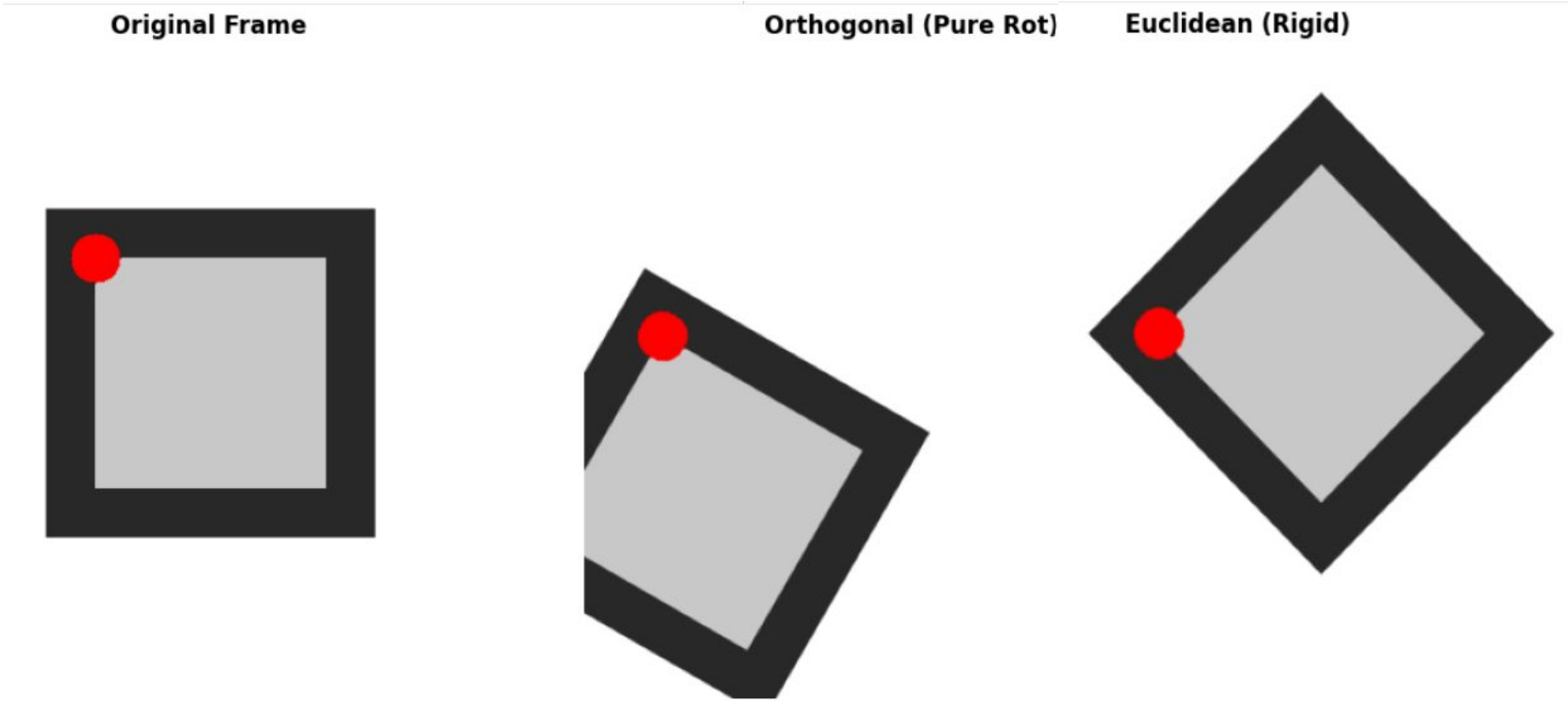
CONTINUATION

```
# ========================================
# PLOT AND DISPLAY RESULTS
# ========================================
titles = ['Original Frame', 'Orthogonal (Pure Rot)', 'Euclidean (Rigid)', 'Affine (Shear/Scale)', 'Projective (Perspective)']
images = [img, img_orthogonal, img_euclidean, img_affine, img_projective]

plt.figure(figsize=(15, 10))
for i in range(5):
    plt.subplot(2, 3, i + 1)
    plt.imshow(cv2.cvtColor(images[i], cv2.COLOR_BGR2RGB))
    plt.title(titles[i], fontsize=12, fontweight='bold')
    plt.axis('off')

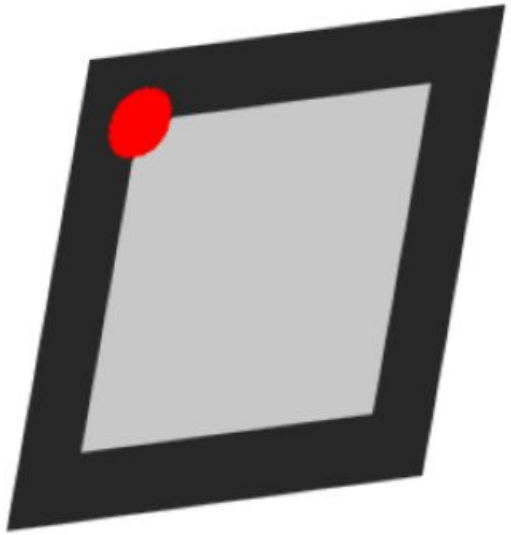
plt.tight_layout()
plt.show()
```

CODE OUTPUT

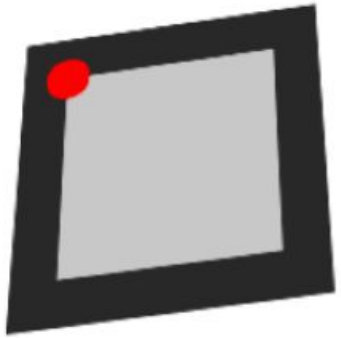


CONTINUATION

Affine (Shear/Scale)



Projective (Perspective)



ADVANTAGES

1. **Improved Accuracy**: Digital imaging is less susceptible to human factors and gives accurate output of the object with high detailed capture.
2. **Enhanced Flexibility**: Digital images are easy to manipulate, edit or analyse as per the requirements through different software hence they provide flexibility of post processing.
3. **High Storage Capacity**: Data in any digital format such as in one or more digital images can still be stored in large amount with very high resolution and quality and will not suffer physical wear and tear.
4. **Easy Sharing and Distribution**: The use of digital images allows them to be quickly duplicated and transmitted across various channels and to various gadgets, helping to speed up the work.
5. **Advanced Analysis** Capabilities: Digital imaging enables the application of analytical tools, including image recognition and machine learning, which can provide better insights and increase productivity.

DISADVANTAGES

1. **Data Size:** Large-structured digital image could occupy large storage space and computational power hence may be expensive.
2. **Image Noise:** Digital images may be compromised by noise and artifacts, which degrades the image quality mainly when photographed at night or using low image sensors.
3. **Dependency on Technology:** Digital imaging entails the use of sophisticated technology and equipment that may be costly and there may be constant need to service or replace the equipment.
4. **Privacy Concerns:** The ability to take and circulate photographs digitally also poses concern because personal information can be photographed without the subject's permission.
5. **Data Loss Risks:** Digital image repositories, however, are prone to data loss caused by hardware failures, corrupting software, or unintentional erasure.

APPLICATIONS

1. **Medical Imaging**: Digital imaging is employed in the medical fields in the diagnostic process such as X-ray pictures, MRI scans, and CT scans, for internal body reflections.
2. **Surveillance and Security**: Digital cameras and imaging systems are greatly needed for various security or surveillance purposes as they offer live feed and are also useful in acquiring data for investigations.
3. **Remote Sensing**: Digital imaging plays an important role in remote sensing applications in terms of monitoring and mapping of environment and disasters and involve data captured from satellite and aerial systems.
4. **Entertainment and Media**: The entertainment industry involves the use of digital imaging in films, video games, and virtual reality to deliver improved visual impact.
5. **Scientific Research**: Digital imaging helps in scientific studies through providing best picture at research fields like astronomy, biology, and material science.

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